

Multiple Choice (10 marks)

1. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging
2. The stack is memory for storing
 - (a) Local variables
 - (b) Program code
 - (c) Dynamically linked libraries
 - (d) Global variables
3. What is penetration testing?
 - (a) A procedure for testing libraries or other program components for vulnerabilities
 - (b) Whole-system testing for security flaws and bugs
 - (c) A security-minded form of unit testing that applies early in the development process
 - (d) All of the above
4. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
 - (a) No, I cannot compute the key.
 - (b) Yes, the key is $k = m \text{ xor } c$.
 - (c) I can only compute half the bits of the key.
 - (d) Yes, the key is $k = m \text{ xor } m$.
5. Which among them has the strongest wireless security?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3

True / False (10 marks)

Answer True or False

6. True or False: Troya is classified as a remote Trojan that can control the infected system from a distance.
7. True or False: A Base Signal Station is the primary component used by mobile operators for enhancing signal coverage in cellular networks.
8. True or False: The Tor browser is essential for accessing the Tor network and using its various services anonymously.
9. True or False: The Heartbleed bug allows for reading data outside the bounds of a buffer, potentially exposing sensitive information in memory.
10. True or False: The Morris Worm, released in 1988, is considered the first recognized buffer overflow attack in cybersecurity history.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss why traditional antivirus scanners are ineffective in detecting vulnerabilities like Heartbleed, focusing on the nature of the exploit and the limitations of signature-based detection methods.
12. Discuss the role of address sanitizer (ASAN) in the fuzzing process for detecting memory errors in software. How does ASAN enhance the identification and analysis of these errors?

End of Paper